

Marginal PMF and Covariance



L18_final.pdf
PDF

Generate a lecture scribe suitable for exam preparation for this lecture. While writing the scribe, reason step by step through definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce material that is not present in the context. The output must be a lecture scribe intended as exam reference material. Prioritize correctness, include explicit reasoning when appropriate, follow the lecture's structure and scope, and do not invent content. A lecture scribe is a faithful, exam-ready reconstruction of what was taught in class, written so that a student can later rely on it as reference material while studying. The scribe should function as reference material suitable for preparing for a reading-based / closed notes exam. The context is fixed and authoritative and everything included must be traceable to the provided context.

For this lecture, which focuses on a set of related probability concepts and their applications, write the scribe strictly following the flow of the slides and include all definitions, notation, assumptions, results, derivations, and worked examples exactly as they appear. Wherever there are proofs or derivations, write them step by step so that the reasoning is clear and complete and can be followed during revision. If the lecture involves probability distributions, expectations, conditional probabilities, transformations, or any structured methods, include all formulas and intermediate steps properly and connect them logically. If there are any procedures or structured steps, present them clearly in order without changing their structure. Pay attention to how the different concepts introduced in this lecture build on each other, and make those dependencies explicit so that the progression of ideas is clear. Do not simplify anything or add extra intuition beyond what is given in the lecture. Do not introduce new examples or alternative methods. Write it in a structured way with clear sections so that it reads like proper revision notes, and such that if I study only this document, I should be able to recall the concepts, reconstruct the reasoning, and solve similar problems in an exam.

Here is your **exam-ready lecture scribe for Lecture 18** based strictly on the provided slides.

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

1.1 Definition

For discrete random variables X and Y , the **joint PMF** is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMFs** are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable** (Page 6).

1.2 Marginal PMF from Joint PMF Table

From a joint PMF table:

- Row sums $\rightarrow p_X(x)$
- Column sums $\rightarrow p_Y(y)$

Formally:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

1.3 Example

Given joint PMF:

	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Step-by-step solution

Compute $p_X(x)$:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Compute $p_Y(y)$:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Answer

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

2. Correlation

2.1 Definition

A basic way to measure joint behavior is through the **product moment**:

$$R_{XY} = E[XY]$$

- It represents the **average value of the product XY** (Page 12).
 - It is called a **raw correlation measure**.
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2.2 Interpretation and Limitation

- R_{XY} depends on the **scale of X and Y** .
 - It is computed **about the origin**, not about means.
 - It mixes:
 - Mean values
 - Spread
 - Dependence
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Important Result

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

Transition

To remove mean effects $\rightarrow$ **center variables** $\rightarrow$ **covariance**

3. Covariance

3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation

- Measures how **centered variables vary together**
 - Captures whether deviations:
 - Move in same direction → positive
 - Move in opposite direction → negative
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3.2 Interpretation Summary (Page 15)

- $\text{Cov}(X, Y) > 0$: positive covariance
- $\text{Cov}(X, Y) < 0$: negative covariance
- $\text{Cov}(X, Y) = 0$: no linear co-variation

Important:

- Indicates **direction only**
 - Magnitude depends on **units** → **not reliable for strength**
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3.3 Algebraic Form (Derivation)

Start with:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Step 1: Expand

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X \mu_Y]$$

Step 2: Apply linearity of expectation

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Step 3: Substitute means

$$= E[XY] - \mu_X \mu_Y$$

Final Result

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

3.4 Properties

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as special case

$$\text{Cov}(X, X) = \text{Var}(X)$$

3. Linearity

$$\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$$

4. Independence

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

⚠ **Important:**

Zero covariance $\Rightarrow$ **uncorrelated**, but **does NOT imply independence** (Page 23)

3.5 Example 1

Given:

$$Y = -2X + 3$$

Find: $\text{Cov}(X, Y)$

Step 1: Use formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Compute XY

$$\begin{aligned} XY &= X(-2X + 3) = -2X^2 + 3X \\ E[XY] &= E[-2X^2 + 3X] = -2E[X^2] + 3E[X] \end{aligned}$$

Step 3: Compute $E[Y]$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 4: Substitute

$$\begin{aligned} \text{Cov}(X, Y) &= (-2E[X^2] + 3E[X]) - E[X](-2E[X] + 3) \\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \\ &= -2E[X^2] + 2(E[X])^2 \end{aligned}$$

Step 5: Use variance identity

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ \Rightarrow \text{Cov}(X, Y) &= -2\text{Var}(X) \end{aligned}$$

4. Pearson's Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

4.2 Properties

- Measures **strength + direction** of linear relationship
- **Normalized covariance**
- **Unit-free**

$$-1 \leq \rho_{XY} \leq 1$$

4.3 Interpretation

- $\rho_{XY} > 0$: positive linear trend
 - $\rho_{XY} < 0$: negative linear trend
 - $\rho_{XY} = 0$: no linear correlation
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5. Example 2 (Continuous Case)

Given:

$$f_{XY}(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

5.1 Compute $E[X]$

$$\begin{aligned} E[X] &= \int_0^4 x f_X(x) dx = \int_0^4 x \cdot \frac{x}{8} dx = \frac{1}{8} \int_0^4 x^2 dx \\ &= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{8} \cdot \frac{64}{3} = \frac{8}{3} \end{aligned}$$

5.2 Compute $E[Y]$

$$\begin{aligned} E[Y] &= \int_1^5 y \cdot \frac{y}{12} dy = \frac{1}{12} \int_1^5 y^2 dy \\ &= \frac{1}{12} \left[\frac{y^3}{3} \right]_1^5 = \frac{31}{9} \end{aligned}$$

5.3 Compute $E[XY]$

Since X and Y are independent:

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

5.4 Compute $E[2X + 3Y]$

$$\begin{aligned} E[2X + 3Y] &= 2E[X] + 3E[Y] \\ &= 2 \cdot \frac{8}{3} + 3 \cdot \frac{31}{9} = \frac{16}{3} + \frac{31}{3} = \frac{47}{3} \end{aligned}$$

5.5 Compute $\text{Var}(X)$

$$E[X^2] = \frac{1}{8} \int_0^4 x^3 dx = \frac{1}{8} \cdot 64 = 8$$

$$\text{Var}(X) = 8 - \left(\frac{8}{3}\right)^2 = \frac{8}{9}$$

5.6 Compute $\text{Var}(Y)$

$$E[Y^2] = \frac{1}{12} \int_1^5 y^3 dy = 13$$

$$\text{Var}(Y) = 13 - \left(\frac{31}{9}\right)^2 = \frac{92}{81}$$

5.7 Compute Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

5.8 Compute Correlation

$$\rho_{XY} = 0$$

6. Jointly Gaussian Random Variables

6.1 Definition

A pair (X, Y) is **jointly Gaussian** if:

- Joint density has **bivariate Gaussian form**
 - Marginals are Gaussian (Page 50)
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6.2 Applications

- Signal processing
 - Communication systems
 - Modeling noisy measurements
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6.3 Joint Gaussian PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where:

- μ_X, μ_Y : means
 - σ_X, σ_Y : standard deviations
 - ρ : Pearson correlation
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6.4 Interpretation

- $\rho > 0$: variables increase together
- Example (Page 53–54):

- $\text{CO}_2 \uparrow \rightarrow \text{Temperature} \uparrow$
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Final Key Takeaways

1. Marginal PMF = sum over joint PMF
 2. Correlation $E[XY]$ is raw and not centered
 3. Covariance removes mean effect
 4. $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
 5. Zero covariance $\neq$ independence
 6. Pearson coefficient normalizes covariance
 7. Joint Gaussian fully described by means, variances, and correlation
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If you want, I can convert this into a **1-page revision sheet** or give you **important exam questions from this lecture**.